

DIANA SOLANO - OROPEZA

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Education

Cornell University

Ithaca, NY

Ph.D in Astronomy

Aug 2023 - Present

- Concentrations: Astrostatistics, Astrobiology, Exoplanets
- Coursework: Exoplanet Characterization, Physics of the Planets, Planetary Atmospheres, Galactic Structure and Stellar Dynamics, Physics of Stars, Neutron Stars, and Black Holes, Geochemistry, Astrophysical Processes, Geodynamics, Bayesian Data Analysis: Principles and Practice
- Thesis Committee: Lisa Kaltenegger (*chair*), Jake Turner, Anna Ho, and Megan Holycross

Columbia University

New York, NY

Post-Baccalaureate Studies in Astronomy and Astrophysics

Sept 2021 - Aug 2023

- Post-Baccalaureate Advisor: David Kipping

Drexel University

Philadelphia, PA

B.S. in Physics, Concentration in Astrophysics | Minor in Sociology

Sept 2016 – June 2021

- Senior Thesis: "Unmixing a Document: Resonance and Language Generation" advised by Jake Ryland Williams

Honors and Awards

NASA Astrobiology Research Poster Finalist

May 2026

2nd Place in Fiction, 2026 Astrobiology Science Conference Creative Writing Contest

May 2026

Cornell Dean's Excellence Fellowship

Mar 2023

National Science Foundation Graduate Research Fellowship

Mar 2023

AAS 241st Meeting FAMOUS Travel Grant

Nov 2022

AAS CSMA Micro-Grant

Nov 2022

Dean's List

Fall 2020, Winter – Spring 2021

A.J. Drexel Scholarship

Fall 2016 – 2020

STAR Scholar

Summer 2017

3rd Place Presentation in Physical Sciences, Greater Philadelphia AMP Conference

Oct 2017

New York City Council Certificate of Recognition

Aug 2016

Research Experience

Astronomy Graduate Research Fellow

Ithaca, NY

Carl Sagan Institute

Sept 2023 – Present

- Expand known stars and exoplanets that may see Earth transit the Sun within the past and future 500,000 years using Gaia satellite data, advised by Lisa Kaltenegger at the Carl Sagan Institute (CSI) and Jackie Faherty at the American Museum of Natural History
- Utilize stellar kinematic and Galactic dynamic models to calculate the motions of stars through millions of years; convert between Galactic, heliocentric, and geocentric coordinate systems; determine time-evolving uncertainties with analytic and stochastic statistical models; estimate habitability of planets with polynomial models to determine high-priority targets for technosignature and biosignature surveys
- Lead radio Search for Extraterrestrial Intelligence (SETI) observation campaign of high-priority targets that have seen Earth transit, with the Allen Telescope Array at the Hat Creek Radio Observatory, in collaboration with Sofia Sheikh and Joe Bright at the SETI Institute, advised by Jake Turner and Lisa Kaltenegger at CSI

- Write bash and Python scripts to manage high-performing computing (HPC) cluster workflows for numeric computation, Nvidia GPU-accelerated inference, and signal processing
- Funded by Cornell Dean's Excellence Fellowship, NSF GRFP, and the Brinson Foundation

Astrophysics Research Assistant

New York, NY

Columbia Astrophysics Labs

Sept 2021 – Aug 2023

- Led exoplanetary science research project, advised by David Kipping in Cool Worlds Lab in Columbia Dept of Astronomy
- Led and trained team of 3 undergraduates, assisted by graduate student Daniel Yahalomi, in labeling light curve data needed to estimate eccentricity
- Estimated eccentricity for approx. 1000 exoplanets orbiting late M stars using TESS transit data, without the need for radial velocity data, using Bayesian statistics
- Produced catalog of eccentricity data to be made available to the general public, computed using NASA Pleiades supercomputer

Applied Physics Research Assistant

Philadelphia, PA

Drexel University College of Computing and Informatics

Sept 2020 – June 2021

- Culminated in senior physics thesis "Unmixing a Document - Resonance and Language Generation", advised by Dr. Jake Ryland Williams
- Stochastic language modeling with aim of demonstrating key physical mechanisms behind human-generated written language
- Verified proposed mechanisms by analyzing data obtained by counting up words in Project Gutenberg corpus, as well as examining applications in quantitative social sciences and machine learning

Data Analyst

Philadelphia, PA

Drexel University College of Computing and Informatics

Aug 2020 – Aug 2021

- NSF REU Supplement Award Number 1850014, supervised by Dr. Jake Ryland Williams
- Studied online human discourse with aim of providing moderation tool against ad hominem attacks in social media
- Stochastic language modeling combined with software development
- Annotated and processed about 500 social media threads for preparation in machine learning tasks

Applied Physics Research Assistant

Philadelphia, PA

Drexel University College of Computing and Informatics

April 2020 – Sept 2020

- Remote co-op funded by Drexel's Alliance for Minority Participation program, supervised by Dr. Jake Ryland Williams
- Studied and supported collaborative projects on stochastic language modeling
- Developed visualizations of probabilistic models and data clusters based on Project Gutenberg corpus composed of 1000 documents after running numerical experiments on cluster computer
- Resulted in preprint paper posted on arXiv

Astrophysics Research Assistant

Philadelphia, PA

Drexel University Department of Physics

June 2017 – Sept 2017

- Part of STAR (Scholars Tackling Advanced Research) summer undergraduate program at Drexel
- Produced mock observations of simulated star formation under supervision of Dr. Stephen McMillan
- Streamlined the process of graphical simulation image production initially developed by graduate student Joshua Wall
- Presented at the Oct 2017 Greater Philadelphia AMP Conference and won 3rd Place in Physical Sciences

Astrophysics Research Assistant

New York, NY

American Museum of Natural History

Sept 2015 – June 2016

- As part of Science Research Mentoring Program, studied dynamics of chaotic binary star system interactions under supervision of museum astrophysicist Dr. Nathan Leigh
- Developed plots of system orbits and virial ratios using Supermongo
- Published a paper in the *Monthly Notices of the Royal Astronomical Society*

Teaching Experience

Special Events and Exhibition Hall Educator

Academy of Natural Sciences

Philadelphia, PA
Oct 2019 – Mar 2020

- Educated visitors on issues and discoveries in the natural sciences using the seasonally rotating special exhibition as a launchpad
- Communicated science concepts such as evolution and the prevalence of living dinosaurs today to museum patrons of all ages at special events using live demonstrations
- Created several public outreach activities such as biodiversity-demonstrating scavenger hunts and collaborated with museum staff to manage events

Student STEM Middle School Teaching Assistant

Drexel University

Philadelphia, PA
Jan 2019 – Mar 2019

- Part of DragonsTeach Step One course offered at Drexel
- Created and taught several lessons pertaining to math topics such as geometry and probability at Masterman School
- Collaborated with fellow teaching student on teaching and managing classroom activities during lessons using research-based education techniques

Student Elementary School Science Club Teaching Assistant

Drexel University

Philadelphia, PA
Jan 2018 – Mar 2018

- Part of Connections in Physics course offered at Drexel
- Collaborated with fellow physics students to manage and run an after-school science club at Alaine Locke School
- Created enrichment lessons that focused on physics topics appropriate for an elementary school audience

Publications

The democratic detrender: Ensemble-Based Removal of the Nuisance Signal in Stellar Time-Series Photometry. —Yahalomi, D., Kipping, D., Solano-Oropeza, D., Li, M., Poddar, A., Zhang, X., Abaakil, Y., Cassese, B., Lennings, J., Larsen, S.S., Turner, J.D., Teachey, A., Sundai, F., Valaskovic, L. (2026) *The Astrophysical Journal Supplement Series*.
doi:10.3847/1538-4365/ae43c2

Near-circular orbits for planets with Earth-like sizes and instellations around M and K dwarf stars. —Kipping, D., Solano-Oropeza, D., Yahalomi, D., Poddar, A., Li, M., Zhang, X. (2025) *Nature Astronomy*. doi:10.1038/s41550-025-02532-8

The chaotic four-body problem in Newtonian gravity I: Identical point particles. —Leigh, N.W.C., Stone, N.C., Geller, A.M., Shara, M.M., Thomas, Y., Solano-Oropeza, D., Muddu, H. (2016) *The Monthly Notices of The Royal Astronomical Society*. doi: 101093/mnras/stw2178.

Manuscripts In Preparation

Tracking exoplanet host stars through the past and future 500,000 years.

—Solano-Oropeza, D., Kaltenegger, L., Faherty, J.K. (2026) In prep.

Closely approaching stars witness to Earth-Sun transits throughout millions of years across the Galaxy. —Solano-Oropeza, D., Kaltenegger, L., Faherty, J.K., Bailer-Jones, C. (2026) In prep.

Who Watches the Watchmen: A SETI radio search of habitable zone transiting planets that have seen Earth transit —Solano-Oropeza, D., Turner, J.D., Sheikh, S.Z., Bright, J., Kaltenegger, L., Faherty, J.K., Jacobson-Bell, B., Lowry, G., Gutierrez, G., Scharfstein, N., Zelakiewicz, A., Larsen, S.S., DeMarines, J., Bibas, D., de Paulis, D., Mastrogiovanni, A., Madden, J. (2026) In prep.

Presentations

Through the Rear Window: Characterizing the Stars that See Us Transit —*Solano-Oropeza, D. (2026)* New York Area Exoplanet Meeting III.

Stars That Can See Earth Transit —*Solano-Oropeza, D. (2026)* International Astronomical Union Symposium 404: Advancing the Search for Technosignatures.

Who Watches the Watchers? Exoplanet Host Stars in the ETZ Throughout Time —*Solano-Oropeza, D. (2025)* 2025 Penn State SETI Symposium.

Welcome to the ETZ —*Solano-Oropeza, D. (2025)* exoNYC 2.

Psst, Are the Neighbors Watching? Stars That Can See Us —*Solano-Oropeza, D. (2024)* Emerging Researchers in Exoplanet Science Symposium (ERES) IX.

Eccentric Exo-Neighbors Orbiting Late M Stars —*Solano-Oropeza, D. (2022)* Columbia University Astrofest 2022.

Estimating Eccentricities for M-Dwarfs without Radial Velocity POP —*Solano-Oropeza, D. (2022)* 2022 Sagan Exoplanet Summer Hybrid Workshop: Exoplanet Science in the Gaia Era.

Mapping the Eccentricities of Pale Red Dots —*Solano-Oropeza, D. (2022)* 2022 Annual Bridge to the PhD. in STEM Research Symposium.

The Mixing Law —*Williams, J.R., Solano-Oropeza, D. (2021)* Fourth Northeast Regional Conference on Complex Systems.

Capturing Chaos in Gravitational Interactions —*Solano-Oropeza, D., Thomas, Y., Muddu, H., Leigh, N. (2016)* 2016 NYC Science Student Research Colloquium.

Posters

Habitable Zone Exoplanets That Could Witness Earth-Sun Transits: ETZ Targets Through 1 Million Years—*Solano-Oropeza, D., Kaltenegger, L., Faherty, J. (2026)* 2026 Astrobiology Science Conference.

Chasing exoplanet host stars through the Earth Transit Zone over the past and future 500,000 years.—*Solano-Oropeza, D., Kaltenegger, L., Faherty, J. (2026)* International Astronomical Union Symposium 404: Advancing the Search for Technosignatures.

Who Watches the Watchmen? Exploring the ETZ with Gaia DR3 —*Solano-Oropeza, D., Kaltenegger, L., Faherty, J. (2024)* Astrobiology Graduate Conference (AbGradCon) 2024.

Probing the Eccentricities for Long-Period Planetary Candidates Orbiting Low-Mass Stars Observed by TESS —*Solano-Oropeza, D., Kipping, D., Yahalomi, D., Poddar, A., Li, M., Zhang, A. (2023)* Columbia University Pathways Program Symposium 2023.

TESS, M-Dwarfs, and the Photoeccentric Effect: A Loophole in Transit —*Solano-Oropeza, D., Kipping, D., Yahalomi, D., Poddar, A., Li, M., Zhang, A. (2023)* AAS 241st Annual Meeting of the American Astronomical Society.

Estimating Eccentricity for M-Dwarfs without Radial Velocity —*Solano-Oropeza, D., Kipping, D., Yahalomi, D., Poddar, A., Li, M., Zhang, A. (2022)* 2022 Sagan Exoplanet Summer Hybrid Workshop: Exoplanet Science in the Gaia Era.

Mapping the Eccentricity of Pale Red Dots —*Solano-Oropeza, D., Kipping, D. (2021)* Simons Foundation Center for Computational Astrophysics GothamFest 2021.

The Mixing Law and Experiments in Document Malformation —*Williams, J.R., Solano-Oropeza, D. (2021)* Fourth Northeast Regional Conference on Complex Systems.

Mock Observations of Simulated Star Formation —Solano-Oropeza, D., Wall, J., McMillan, S. (2017) STAR Summer Showcase 2017.

Capturing Chaos: On the Chaotic 4-Body Problem in Newtonian Gravity —Muddu, H., Solano-Oropeza, D., Thomas, Y., Leigh, N. (2016) 2016 NYC Science Student Research Colloquium.

Preprints

A general solution to the preferential model —Williams, J.R., Solano-Oropeza, D., Hunsberger, J.R. (2020) arXiv:2008.02885

Science Communication Presentations

What if we found an alien? My live-action roleplay experience —Solano-Oropeza, D. (2026) Astronomy on Tap Ithaca.

What is the Galactic Habitable Zone? —Solano-Oropeza, D. (2025) Astronomy on Tap Ithaca.

Speedrunning, Childhood Trauma, and Planetary Transits —Solano-Oropeza, D. (2022) The Science of Everyday Life, A Comedy/Storytelling Workshop at the American Museum of Natural History.

Science Communication Articles

Super-Earth, Super-Tectonics?—Solano-Oropeza, D. (1 Oct 2025) Astrobites.

Smooons, Ploonets, and Circumbinary Planets, Oh My!—Solano-Oropeza, D. (8 Jan 2025) Astrobites.

Giving thanks to Carl Sagan for 90 years of science inspiration —Solano-Oropeza, D. (28 Nov 2024) Astrobites.

It's An Eyeball Summer, and Other Weird K/M-Dwarf Habitable Climate Tales —Solano-Oropeza, D. (15 August 2024) Astrobites.

Noclip on! Simulated primordial black holes could dance through Sun-like stars —Solano-Oropeza, D. (29 May 2024) Astrobites.

Vulcan II: The Wrath of Stellar Activity —Solano-Oropeza, D. (15 May 2024) Astrobites.

How many astrophysicists does it take to catch a RATT? —Solano-Oropeza, D. (22 Feb 2024) Astrobites.

Media Communications

Quoted in **Astronomy professor David Kipping part of breakthrough exomoon discovery** —Pagkas, S., Rossi, E. (2 Feb 2022) The Columbia Daily Spectator.

Leadership

Vice President

Astronomy Graduate Network

Contract Action Team Organizer

Cornell Graduate Student Union

Ithaca, NY

June 2025 – Present

Ithaca, NY

Mar 2024 – May 2025

Skills

Programming Languages: Python, C++ , Mathematica, ADQL, SQL, IDL, Fortran, R

Human Languages: Spanish (Fluent), Tezoatlan Mixteco (Beginner), Classical Nahuatl (Beginner), Portuguese (Beginner), Mandarin Chinese (Beginner)

Tools: Jupyter Notebooks, Git, Supermongo, LaTeX, Microsoft Word/Powerpoint/Excel, Photoshop, Active Directory, Windows, Mac, Linux/Ubuntu, NASA Pleiades, emacs, bash, zsh, Visual Studio Code

Libraries: pandas, matplotlib, numpy, spaCy, astropy, yt, FLASH, RADMC, AMUSE, MESA, exoplanet, orvara, lightkurve